

FIRMWARE BUILD ENVIRONMENT ISSUES & RESOLUTIONS REPORT

Band01_Rev1 Smartband · Task 3 Development Log

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Environment: CCS v12.8.1 · TI CC2640R2 SDK v5.30.01.11 · Windows 11

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BUILD ENVIRONMENT — Context & Overview

Project Context

Target: CC2640R2FRSMR BLE SoC (Smartband V001)

Tool: Code Composer Studio v12.8.1

SDK: TI SimpleLink CC2640R2 SDK v5.30.01.11

App workspace: workspace_v12

Stack workspace: workspace_v13

Goal: Get the firmware to compile and link on a fresh machine

3 Issues — All Resolved

1

Missing build_config.opt

All 25+ source files failed — compiler options file absent from repo

2

Missing BLE Stack Library

Linker could not find the 3 required binary stack files

3

Header Mismatch gapbondmgr.h

Header declared 2 functions differently from compiled library

✓ All 3 issues resolved · 0 errors · Build completed 28 May 2026

ISSUE 1 — Missing build_config.opt

✓ RESOLVED

Error

```
Cannot open command file
'/TOOLS/build_config.opt'
No such file or directory

Repeated for ALL 25+ source
files.

Secondary error:
#error "RF_SINGLEMODE
missing" at line 64 of
RFCC26XX_singleMode.c
```

Root Cause

```
CCS project references:
--cmd-
file="${REF_PROJECT_1_LOC}
/TOOLS/build_config.opt"

REF_PROJECT_1_LOC pointed to
the stack project folder –
never committed to GitHub.

On a fresh machine this
resolves to /TOOLS/... – a
path that doesn't exist.
```

Resolution

```
Manually created:
C:\TOOLS\build_config.opt

With 8 required preprocessor
flags:
-DRF_SINGLEMODE
-DUSE_ICALL
-DICALL_LITE
-DPOWER_SAVING
-DHEAPMGR_SIZE=0
-DMAX_NUM_BLE_CONNS=1
...

Committed to repo at
TOOLS/build_config.opt
```

ISSUE 2 — Missing BLE Stack Library

✓ RESOLVED

Linker Errors:

```
#10008-D cannot find "/FlashROM_Library/.lib"  
#10008-D cannot find  
"/FlashROM_Library/ble_r2.symbols"  
#10008-D cannot find  
"/FlashROM_Library/lib_linker.cmd"  
#10192 Failed linktime optimization  
symbol "RF_cancelCmd" redeclared... (15+ more)
```

Root Cause:

The 3 files are compiled outputs of the BLE stack library project (smart_band_v001_stack_library) — never committed to repo (binary files).

First attempt used BLE5 stack by mistake → 20+ new symbol conflicts. Correct stack: blestack (BLE4), not ble5stack.

Resolution — 3 Files Required:

.lib

Stack library binary
→ C:\FlashROM_Library\.lib

lib_linker.cmd

Linker command file
→ C:\FlashROM_Library\
lib_linker.cmd

ble_r2.symbols

⚠ These are binary files — cannot be committed to the repo.
Every new developer must build the BLE4 stack library from SDK blestack examples and copy the 3 files to C:\FlashROM_Library\

ISSUE 3 — gapbondmgr.h Header Mismatch

✓ RESOLVED

Final Linker Errors:

```
symbol "GAPBondMgr_PasscodeRsp" redeclared with incompatible type: smart_band_v001_app
symbol "GAPBondMgr_Register" redeclared with incompatible type: smart_band_v001_app
#10192 Failed linktime optimization
```

The Problem — Header vs Library Version Mismatch:

Include/gapbondmgr.h (repo copy)

OLD version — declares GAPBondMgr_PasscodeRsp and GAPBondMgr_Register with outdated parameter types

≠

Compiled .lib (from current SDK)

NEW version — same functions compiled with updated signatures. Linker detects conflict and refuses to link.

Resolution:

Replaced **Include/gapbondmgr.h** with the correct version from:
C:\ti\simplelink_cc2640r2_sdk_5_30_01_11\source\ti\blestack\profiles\roles\gapbondmgr.h
File committed back to repository so future developers get the correct header on clone.

FINAL BUILD RESULT — 0 Errors

✓ **smart_band_v001_app — BUILD SUCCESSFUL — 0 Errors — 28 May 2026**

All 3 issues resolved in sequence · Output files generated · Ready to flash via JTAG

Output Files Generated:

smart_band_v001_app.out

ELF Binary

Used by CCS debugger for flashing and step-through debugging via JTAG probe

smart_band_v001_app.hex

Intel HEX

Used for direct flashing tools — stored in repository from original build

smart_band_v001_app.map

Linker Map

Shows exact Flash and RAM memory usage for every compiled module

Firmware state: Sensor calls are currently commented out — transmits placeholder value **0x7788** via BLE GATT. Ready to flash once JTAG probe and power supply are connected.

FUTURE DEVELOPER SETUP GUIDE

1

Clone repository

TOOLS/build_config.opt and Include/gapbondmgr.h are now present — no manual recreation needed

2

Install TI SDK

TI SimpleLink CC2640R2 SDK v5.30.01.11 required at
C:\ti\simplelink_cc2640r2_sdk_5_30_01_11\

3

Import & build stack library

Import simple_peripheral_cc2640r2lp_stack_library from
blestack SDK examples into a separate workspace
(workspace_v13)

4

Copy 3 output files

Copy .lib, lib_linker.cmd, ble_r2.symbols from build output to
C:\FlashROM_Library\

5

Build app project

Import smart_band_v001_app into workspace_v12 and build
— should complete with 0 errors

6

Flash via JTAG

Connect XDS110 debug probe, power the board, use CCS to
flash smart_band_v001_app.out

SUMMARY

Issue 1

build_config.opt was absent from the repo — created and committed

Issue 2

BLE4 stack library missing — built from SDK blestack examples, placed at C:\FlashROM\

Issue 3

gapbondmgr.h out of sync with SDK — replaced with correct version and committed

Result

0 errors · Build completed 28 May 2026 · .out, .hex, .map files generated

Firmware

Transmits placeholder 0x7788 via BLE GATT — ready to flash once JTAG probe available